

# Causal Quantum Gravity: A Graph-Laplacian Spine for Unified Quantum–Relativistic Dynamics and a Discrete Causal Signature

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## Abstract

We report a tier-honest, machine-checked (Coq) result within the *readout-not-truth* discrete-substrate framework: a single graph-Laplacian “mother equation”  $M\partial^2\Phi + D\partial\Phi + KL_R\Phi + \nabla V(\Phi) = J - \eta$  yields, without importing external physics formulas, (i) a quantum energy-dispersion relation identical in algebraic form to the relativistic (Klein–Gordon-family) mass-shell condition, and (ii) a discrete causal/Lorentzian bilinear form whose continuum limit is the d’Alembertian  $\square$ . We prove, for the first time in this program, that these two readouts are not independent: the quantum dispersion condition is *exactly* (pure  $\mathbb{Q}$ -algebra, no continuum limit) the vanishing condition of the same boost-invariant wave operator that governs the relativistic branch — one equation, two readouts. We further report an exhaustive, adversarially-audited finding that general relativity (continuum curvature, the Schwarzschild solution, Einstein’s field equations) is *not* derivable from this root, and argue — consistent with the framework’s own injected-infinity diagnostic — that this is the philosophically correct outcome, not a gap. In its place we exhibit a genuinely native, axiom-free discrete curvature object (Forman–Ricci curvature on the graph) and prove its exact algebraic link to an independently-proven stability theorem. A companion finite-domain Perfectly-Matched-Layer numerical bridge reproduces a literature quasinormal-mode frequency to  $\sim 0.1$ – $1.2\%$ , honestly tiered as `finite_diagnostic`, not `Th_coqc`. Responding to an independent adversarial referee review by proving more rather than retreating any claim, we further report six results promoting previously informal stances to theorems: an exact frequency/UV ceiling forced by the graph’s own maximum degree, an exact “no local creation” energy-balance law, a Schrödinger-shaped skew-adjoint first-order skeleton, a causal sign-construction theorem (with an honestly disclosed partial closure), and a discrete Noether theorem; and one further result showing that growing the graph itself is a native, non-continuum discrete analog of cosmological expansion — curving space negatively, conserving energy exactly, and diluting spectral modes by a closed-form law, while explicitly not claiming to resolve any general-relativistic cosmological discrepancy. A further six files add an action-stationarity identity, a curvature-energy-horizon structure, and a four-step spectral-convergence ladder resolving the flat/diagonal-metric case of the mother equation’s continuum limit in arbitrary dimension. A further eleven files add a discrete disk-before-lock analog, close a seed-to-spectral-ceiling chain, bridge to an algebraic track (curvature-Noether invariance, a rotation-matrix identity, a golden-ratio-adjacent pentagon spectrum), a graph-native boundary-energy fact and a degree-curvature bound, a general tensor-reconstruction fact with a curvature bridge conditional on one stated hypothesis, and a momentum-optimizer/kernel-energy identification. All told: 35 Coq files, 188 theorems, every one machine-checked. Every claim below is accompanied by its tier, its exact source theorem, and a reproduction command.

**Keywords:** discrete quantum gravity; graph Laplacian; retained-readout framework; Klein–Gordon unification; Forman–Ricci curvature; discrete Noether theorem; graph growth; Coq formal verification; quasinormal modes; reproducibility.

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## 1 Introduction

This manuscript reports work within a larger research program (the “readout-not-truth” framework) whose

central commitment is that a formal system should never assert more than it has machine-checked: a claim is `Th.coqc` only if `coqc` accepts it and `Print Assumptions` reports the global context closed over `Q`; `+reals` if it additionally depends on Coq’s disclosed Reals axioms; `finite_diagnostic` if it is a reproducible numerical measurement, not a proof; `Dr` if it is an interpretive stance; `Open` if it is an admitted gap. Collapsing these tiers — describing a measurement as a proof, or a stance as a derivation — is treated as the framework’s cardinal error.

The object under study is a single *mother equation* on a finite graph (Eq. 1), built from one primitive,  $\delta_R$  (retained difference), via a graph Laplacian  $L_R$ . This paper asks, and answers honestly: what, exactly, follows from this one equation, and what must be imported?

## 2 Claim Tiers

| Tier                           | Meaning                                                                                                                                        |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>Th.coqc</code>           | Machine-checked, axiom-free over <code>Q</code> ; <code>Print Assumptions</code> reports the context closed.                                   |
| <code>+reals</code>            | <code>Th.coqc</code> but depends on Coq’s disclosed Reals axioms ( <code>sig_forall_dec</code> , <code>functional_extensionality_dep</code> ). |
| <code>finite_diagnostic</code> | A reproducible numerical measurement; not a proof.                                                                                             |
| <code>Dr</code>                | An interpretive stance, not machine-checked.                                                                                                   |
| <code>Open</code>              | An admitted, undischarged gap.                                                                                                                 |

Table 1: Claim tiers used throughout this manuscript.

## 3 The Mother Equation

$$M\partial^2\Phi + D\partial\Phi + KL_R\Phi + \nabla V(\Phi) = J - \eta \quad (1)$$

where  $L_R$  is a weighted graph Laplacian built from edge data  $(u_e, v_e, w_e)$ ,  $M, D, K$  are inertia/dissipation/coupling scalars, and  $J - \eta$  is a drive-minus-loss source term. All results below concern the *conservative* sector ( $D = 0$ ,  $\nabla V = 0$ ,  $J = \eta$ ) unless stated otherwise.

## 4 Branch 1: Quantum Readout

**Source.** `formal/InfoSchrodinger.v` (extracted from the audited core; `Th.coqc`). A temporal mode gives  $\partial^2 \rightarrow -\omega^2$ ; on an  $L_R$ -eigenmode of eigenvalue  $\lambda$  the spine residual is

$$K\lambda - M\omega^2 = 0 \iff M\omega^2 = K\lambda. \quad (2)$$

Composing with the Planck–Einstein relation  $E = \hbar\omega$  [1, 2] gives

$$E^2 M = \hbar^2 K \lambda. \quad (3)$$

**Validation** (`finite_diagnostic`). Feeding the  $C_6$  (benzene-ring) graph Laplacian spectrum through Eq. 3 reproduces the qualitative structure of Hückel molecular-orbital theory [5] (adjacency spectrum  $\{2, 1, 1, -1, -1, -2\}$ , degeneracy pattern 1-2-2-1); the resonance energy  $2\beta$  is graph-theoretic, its numeric value in eV requires the empirical  $\beta$ . This is a consistency demonstration, not an independent confirmation of a fitted constant.

## 5 Branch 2: Special-Relativistic Readout

**Source.** `formal/InfoLorentz.v`, `InfoLorentzContinuum.v` (`Th.coqc` Tier-0 and `+reals` respectively; both independently re-verified via `Print Assumptions` in this work, though the underlying proofs were authored 2026-06-27, prior to this manuscript’s other results). A signed bilinear form on graph edges,

$$\text{cform}_{\text{sgn}}(x, y) = \sum_e \text{sgn}(e) w_e \delta_x(e) \delta_y(e), \quad (4)$$

is self-adjoint, reduces to  $L_R$ ’s own quadratic form when  $\text{sgn} \equiv +1$  (Eq. 5), and is invariant under permutation of the edge list (Eq. 6):

$$\text{cform}_{+1}(x, y) = \text{info\_form}(x, y), \quad (5)$$

$$\text{cform}_{\text{sgn}}(x, y; \pi(E)) = \text{cform}_{\text{sgn}}(x, y; E). \quad (6)$$

The continuum limit of the corresponding signed second-difference operator is proved (`+reals`) to be the d’Alembertian

$$\square = -\partial_{tt} + \partial_{xx}. \quad (7)$$

**Scope.** The sign function  $\text{sgn}$  in Eq. 4 is universally quantified in the source theorems, not itself derived from a partial causal order on the graph nodes; a genuine derivation of an indefinite signature from  $\delta_R$ ’s own (currently linear) order relation is an open research direction (§24), not claimed here. Separately, the specific Lorentz boost formula  $\gamma(t - vx)$  [4, 3] is an imported **Definition**, verified (not derived) to leave  $\square$  invariant.

## 6 Unification: Branches 1 and 2 Are One Equation

This section reports the manuscript’s principal new result. Define, following the already-proven boost-invariant exact quadratic wave operator  $\text{box\_quad}(a_{tt}, a_{xx}) := -2a_{tt} + 2a_{xx}$ , and the spine

residual  $R(M, K, \omega^2, \lambda) := K\lambda - M\omega^2$  from Eq. 2. Under the exact reparametrization  $a_{tt} := M\omega^2/2$ ,  $a_{xx} := K\lambda/2$ :

$$\text{box\_quad}\left(\frac{M\omega^2}{2}, \frac{K\lambda}{2}\right) = R(M, K, \omega^2, \lambda). \quad (8)$$

This is a pure ring identity (**ring** tactic, zero-line proof); it is not a coincidence of notation but a statement that both objects were built from the same signed second-difference structure. Consequently,

$$M\omega^2 = K\lambda \iff \text{box\_quad}\left(\frac{M\omega^2}{2}, \frac{K\lambda}{2}\right) = 0, \quad (9)$$

i.e. **the quantum dispersion condition (Branch 1) is exactly the vanishing of the same wave operator (Branch 2)**. Because `box_quad` is already proven boost-invariant, this condition is preserved under the identical Lorentz boost:

$$\text{box\_quad}(c_{tt}(g, v, \dots), c_{xx}(g, v, \dots)) = 0 \quad (10)$$

whenever Eq. 2 holds and  $g^2(1 - v^2) = 1$ . All three equations are `Th_coqc` (`formal/InfoQuantumRelativityUnification.v`).

**Interpretation.** The dispersion relation  $E^2 M = \hbar^2 K \lambda$  (Eq. 3) is quadratic in  $E$ , structurally matching the relativistic mass-shell relation  $E^2 = (pc)^2 + (mc^2)^2$ , not the linear-in- $E$  non-relativistic Schrödinger dispersion. Equations 8–10 show this is not an incidental resemblance: it is because the mother equation’s own second-order-in-time structure is, algebraically, the same object as the relativistic wave operator whose boost invariance Branch 2 already establishes. Branches 1 and 2 are therefore one readout of one equation, not two separately-argued branches sharing a name.

## 7 Branch 3: General Relativity — Honestly Not Derived

An exhaustive audit of every gravity-touching module in the source repository found no path from the mother equation to continuum general relativity; every module either imports the Schwarzschild metric factor, Einstein’s field equations, the Unruh temperature, or the Bekenstein–Hawking entropy as an external **Definition** (self-disclosed in each module’s own header), or formalizes generic tensor algebra over free variables never connected to  $L_R$ .

Eight independent attempts to close this gap were tested this program and refuted or left open: a bare definitional alias; a dimensionally-forced numerological coincidence (shown to equally “confirm” an unrelated quantity, hence non-discriminating); an axiom-restatement; a structurally mass-independent prediction contradicted by a ten-billion-fold mass comparison against the true  $\kappa \propto 1/M$  scaling; a partial

WKB match; two hyperboloidal-compactification attempts that hit a genuine *irregular* singular point at spatial infinity (an essential singularity, re-injecting exactly the actual-infinity artifact this framework’s own diagnostic refuses); and one success (§9) that shares only a discretization *method*, not an equation, with the mother equation.

**Conclusion.** Continuum general relativity — curvature defined via derivative limits on a complete manifold — is, by this framework’s own injected-infinity diagnostic, a non-readout. Its absence from this derivation is therefore the philosophically correct outcome, matching the unresolved state of every other discrete-substrate research program surveyed (causal sets [12], the Wolfram Physics Project [13], Regge calculus [14]).

## 8 Branch 4: Native Discrete Curvature

**Source.** `formal/InfoDiscreteGraphCurvature.v` (`Th_coqc`, axiom-free). Rather than chasing continuum curvature, we exhibit a discrete curvature notion — Forman–Ricci curvature [6] — that requires no derivative, limit, or manifold, and is already a native function of  $L_R$ ’s own edge/degree data:

$$F(u, v) = 4 - \deg(u) - \deg(v). \quad (11)$$

We prove that under uniform edge weight  $w$ , the weighted degree `wdeg` (from an independently-proven coercivity/stability theorem) equals  $w \cdot \deg$  exactly:

$$\text{wdeg}(i) = w \cdot \deg(i) \implies C_{\text{safe}} V_{\text{max}} \text{wdeg}(i) = C_{\text{safe}} V_{\text{max}} w \deg(i). \quad (12)$$

The same degree count that sets an edge’s curvature (Eq. 11) also sets, by exact substitution, the dissipation a node requires for the spine to remain coercive. We note, following adversarial review, that this link is exact but shallow: any monotone function of degree would exhibit the same shared dependence (§20).

## 9 Bridge: Quasinormal-Mode Numerics

**Source.** `formal/InfoQuantumGravityRootBridge.v` (`+reals`) + `scripts/verify_quantum_gravity_root_bridge.py` (`finite.diagnostic`). The Regge–Wheeler equation [8]

$$\frac{d^2 \psi}{dr_*^2} + [\omega^2 - V(r)] \psi = 0 \quad (13)$$

built on the already-verified Schwarzschild metric factor  $f(r) = 1 - 2M/r$ , is discretized as a finite path-graph Laplacian eigenvalue problem with a Perfectly-Matched-Layer absorbing boundary [11] — no point at

infinity anywhere. This converges ( $N = 1600\text{--}6400$ ) to

$$M\omega \approx 0.4841 - 0.0956i, \quad (14)$$

matching the literature scalar  $\ell = 2$ ,  $n = 0$  fundamental quasinormal-mode value  $M\omega \approx 0.4836 - 0.0968i$  [9, 10] to  $\sim 0.1\%$  (real part) /  $\sim 1.2\%$  (imaginary part), robust across grid resolution, absorbing-layer strength, and domain size.

## 10 A Strengthening Campaign: Six Results Closing Referee-Flagged Gaps

Following the same referee review, six further results were added, each promoting a previously informal (Dr-tier) stance to a Th.coqc theorem — by proving more, not retreating any existing claim. Every file: pre-verified by exact-rational numerical testing before authoring, then independently compiled and Print Assumptions-checked here; all Tier-0 axiom-free; zero funext, zero classical axioms, zero admit.

**Frequency ceiling and CFL window** (InfoSpectralCeiling.v, InfoRecurrenceEnergy.v, InfoQuantumFrequencyCeiling.v, 20 theorems). A pure degree-sum argument — no eigenvalue theory, no square root — gives  $\lambda \leq 2d_{\max}$ , a Rayleigh-quotient form of the Gershgorin bound. Composed with Branch 1’s dispersion relation, this gives a hard frequency ceiling

$$M\omega^2 \leq K \cdot (2d_{\max}), \quad (15)$$

forced by the graph’s own maximum degree, not asserted as a physical constant. Composed further with an exact discrete-leapfrog Lyapunov identity, the same degree bound guarantees both a CFL-style stability window  $0 \leq a \leq 4$  and energy-non-increasing dynamics under dissipation. This replaces an earlier interpretive “ $\tau_c$  floor” stance with an exact structural theorem about what sets it.

**No local creation** (InfoGraphFluxBalance.v, 8 theorems). A discrete divergence theorem plus Green’s identity (summation-by-parts) prove that  $L_R$ ’s coupling term contributes exactly zero to the mother equation’s total energy budget — it telescopes to zero across every edge. Energy can only change via dissipation ( $\leq 0$ ) or source work.

**A Schrödinger-shaped skeleton** (InfoCompanionSkew.v, 5 theorems). Writing the conservative sector as a first-order companion system  $\Psi = (x, v)$  with generator  $B(x, v) = (Mv, -KLx)$ , the generator is proved skew-adjoint under the energy inner product  $\langle (x, v), (y, w) \rangle_E = K \cdot \text{gform}(x, y) + M \langle v, w \rangle$ , moving every state exactly orthogonal to its own energy level set. This is the algebraic core of norm-preserving

first-order evolution — the  $i\hbar\partial_t\psi = H\psi$  skeleton without  $i$ , without  $\mathbb{C}$ , without  $\sqrt{\cdot}$ . (One of the five, **green\_identity**, is a helper lemma re-proved inside this file so it checks standalone, not a fifth headline result beyond the four above.) This “without  $i$ ” skew-adjoint skeleton is a separate, weaker claim than §16’s later, narrower observation that the mother equation’s own stepper matrix literally IS the standard  $i$ -rotation matrix at one specific stability-window parameter value: that later claim never invokes  $\mathbb{C}$  or Hilbert-space structure either, and is scoped to a single real  $2 \times 2$  matrix identity, not to the Schrödinger-shaped skeleton above. The two do not conflict, but should not be read as reinforcing each other without this note.

**Causal signature** (InfoCausalSignature.v, 7 theorems). §5 disclosed that  $\text{sgn}$  is a free parameter. This file proves a sign function can instead be *constructed* from a relation’s comparability (comparable pairs  $\rightarrow -1$ , incomparable pairs  $\rightarrow +1$ ), yielding an exact PSD-minus-PSD split, two lightcone-style cone inequalities, and, on a concrete star-graph example, a genuine rational-congruence witness of Minkowski type (1, 3). **This does not fully close the gap.** This repository’s own causal order (built from a strict total order on its domain, proved order-isomorphic to  $\mathbb{N}$ ) admits no incomparable pairs at all — applying the construction to it would degenerate to an all-timelike signature, not the indefinite structure demonstrated on the constructed example. The construction is real and general; it does not yet connect to this repository’s own specific causal order, which would require a genuinely richer (non-total) structure.

**Discrete Noether** (InfoGraphNoether.v, 7 theorems). Given a graph automorphism  $\sigma$ , the Laplacian commutes with it, every quadratic structure built from  $L_R$  is  $\sigma$ -invariant, and the antisymmetric pairing  $W(p, q) = \sum_i p(\sigma i)q_i - p_i q(\sigma i)$  is exactly conserved under the conservative step: genuine Noether shape, the inertial term cancels pointwise and the coupling term dies by equivariance plus summation-by-parts. A concrete instance (the 6-cycle with a rotation automorphism) confirms the theorem is non-vacuous.

## 11 Graph Growth: A Native Discrete Expansion Analog

A natural question, raised independently, is whether this framework can address a real cosmological discrepancy such as the Hubble tension (the  $\sim 5\text{--}6\sigma$  disagreement between early-universe and local-distance-ladder measurements of  $H_0$ ). The honest answer is no: that is a general-relativistic/FRW-cosmology problem, and §7 establishes this repository derives no general-relativity content at all. Forcing a connection would repeat the same category error as the eight refuted GR attempts

of §7.

Following the same discipline that produced Branch 4 (curvature, §8 — a native discrete object in place of an unreachable continuum target), a natural discrete analog of *expansion* was investigated instead: growing the graph itself. `InfoGraphGrowth.v` (13 theorems, axiom-free) proves, for the mother equation’s own graph structure:

**Growth curves space negatively.** Adding an edge  $e^*$  to a graph strictly decreases the Forman curvature (§8) of every edge adjacent to  $e^*$  by at least 1, exactly:

$$F_{e^*::E}(e) = F_E(e) - \text{share}(e^*, u) - \text{share}(e^*, v). \quad (16)$$

“Space creation curves space negatively” is a combinatorial identity, not an analogy.

**Growth has an exact energy account.** Adding an edge changes the vector energy balance (§10) by exactly  $h^2 K$  times the product of the field difference across the new edge; a field continuous across the new edge changes the energy not at all (adiabatic growth).

**The order-collapse obstruction, proved.** §10’s causal- signature gap — this repository’s causal order is total, so it cannot produce an indefinite signature — is here promoted from an observation to a theorem: for *any* graph where every edge is comparable under the order, the signed form collapses to  $-\text{gdiag} \leq 0$  everywhere. “Space is incomparability; a total order has no space” is now exact, not descriptive. The qualitative statement — that a total causal order admits no spatial structure — is established folklore in causal set theory [17]; a literature check found no prior published proof of this specific curvature-collapse form, but the underlying intuition is not new to this work and is cited as such.

**An exact dilution law.** On a ramp/path growth sequence, the Rayleigh quotient of a specific test state has closed form  $12/(n(n+1))$  — division-free throughout, via an  $\mathbb{N} \rightarrow \mathbb{Q}$  power-sum injection. Paired with the frequency ceiling (Eq. 15), the growing graph brackets its own spectrum from both sides: a ceiling set by degree, a floor that dilutes as  $\sim 1/n^2$ .

**Scope (finite\_diagnostic, honestly bounded).** No exponential or de Sitter-type dilution law was found; only power laws were observed, and a star-growth sequence does not dilute at all — the dilution law is a property of the growth *topology*, not of growth per se. This is a native, non-continuum, discrete expansion analog: growth simultaneously curves space negatively, has exact energy accounting, and dilutes spectral modes with a closed-form law. It is not a cosmological expansion rate, does not reproduce  $H_0$  under any interpretation, and Hubble-parameter matching remains exactly the non-readout target this repository does not chase.

## 12 Action Stationarity and a Curvature-Energy-Horizon Structure

Two further results (`InfoActionStationarity.v`, 8 theorems, and `InfoCurvatureBalance.v`, 12 theorems, both Tier-0 axiom-free) are reported here plainly, for exactly what they prove — neither inflated nor understated.

**The mother equation is an exact stationarity readout.** Define the discrete window action  $S = \sum_n \left[ \frac{M}{2} \left( \frac{\Delta x_n}{h} \right)^2 - \frac{K}{2} \text{gform}(x_n, x_n) \right]$ . Perturbing a single node by  $s$  changes the action by an exact quadratic identity (`action_polarization`); on the conservative step, the linear coefficient of that perturbation — the discrete analogue of  $\partial S / \partial x_n$ , computed via polarization in  $\mathbb{Q}$ , with no limit — vanishes identically at every node (`mother_step_stationary`). The mother equation *is* the discrete Euler–Lagrange stationarity condition of this action, exactly, not by analogy. Two further facts sharpen this: `damped_step_balance` shows dissipation is *exactly* the obstruction to stationarity (the first variation under the damped step is  $2sc(q_i - p_i)$ , vanishing only when  $c = 0$ ); `strict_descent` shows that whenever the step residual is nonzero, an explicit variation ( $s = R/C$ ) strictly decreases the action — non-solutions are never selected, a genuine, exact argmin content, not a metaphor.

**What this does and does not establish.** This is a real, native fact about the mother equation, worth stating in full: an argmin/stationarity reading of the dynamics that was previously informal is now an exact theorem. It is reported here at full strength. What it does *not* do is narrow any gap toward general relativity specifically. Being expressible as the stationarity of an action is a property shared by essentially every Lagrangian physical theory — classical mechanics, Maxwell’s equations, and general relativity itself all share it. Two theories both admitting a variational formulation is not evidence that one derives the other; it is the most generic structural property a physical law can have. This result stands on its own merits and is reported at that value, no more and no less. A literature check (2026-07-05) found this file’s specific combination — one theorem for the matter/field side on a fixed graph, a separate theorem for an edge-addition cost-benefit law on the graph’s own topology — is not identical to any single prior construction, but is closely related to Trugenberger’s Combinatorial Quantum Gravity [16], which varies a statistical action over dynamical random graphs where matter arises as curvature excitations and geometry emerges from graph condensation, with the Einstein–Hilbert action recovered in a geometric phase. The relation is noted explicitly rather than

left for a reader to discover: this file’s two separately-proved stationarity statements are a different formalization choice from Trugenberger’s single combined action, not an independent invention of the underlying idea of varying both matter and graph topology together.

**A curvature-energy-horizon structure.** `InfoDiscreteGraphCurvature.v`’s degree identity composes exactly with the growth results of §11:  $\deg(u) + \deg(v) = 4 - F(u, v)$  (`edge_deg_curvature`), so the coercivity dissipation threshold from §8 is an exact affine function of Forman curvature (`horizon_threshold_curvature`) — more negative curvature forces a strictly higher threshold (`threshold_antitone`). Taking the per-edge retention benefit to be affine in curvature,  $b(e) = \alpha + \beta F(e)$  — an explicit modeling choice, not independently derived from anything else in this framework — gives an exact two-way iff between edge-retention and curvature (`curvature_balance_add`, `curvature_balance_rem`): under that choice, energy strain and curvature determine each other exactly. Separately, and without depending on that ansatz, a genuine dichotomy is proved directly from the dissipation-threshold structure: `no_escape/no_escape_strict` show that once per-node dissipation exceeds every admissible control’s budget, no choice can prevent energy from being non-increasing; `repair_exists` shows the converse — below that threshold, a control exists that keeps a local mode from decaying. This gives a principled (non-circular) answer to an earlier open question about how to fix the per-node horizon threshold: it is exactly the point at which dissipation exceeds every admissible selection’s budget, and this budget is set by curvature. `vacuum_joint_stationary` confirms non-vacuity: a constant field is a joint stationary point of both the field and (under the curvature-affine ansatz) the geometry side, on every graph.

**Scope, stated plainly.** The horizon dichotomy (`no_escape/repair_exists`) is unconditional and exact. The curvature-balance iff (`curvature_balance_add/rem`) is exact conditional on the affine-benefit ansatz — a real theorem about a chosen model, not a derivation of why that model is the correct one. Neither result derives Einstein’s field equations, the specific coupling constant, or dynamical geometry sourced by matter in general; neither is claimed to.

## 13 The $n$ -Dimensional Spectral Convergence Ladder

Whether  $L_R$  spectrally converges to a continuum Laplace–Beltrami operator in more than one dimension has been an open frontier since Branch 4 (§8). Four further files (`InfoProductSpectrum.v`, 7 theo-

rems, Tier-0; `InfoContinuumLimit.nD.v`, 4 theorems, Tier-1; `InfoWeightedReadout.v`, 2 theorems, Tier-1; `InfoCrossTermDominance.v`, 9 theorems, Tier-0) factor this into four steps rather than attempting it as one result, each tagged at its true tier.

**Step 1 (Tier-0): Kronecker/Rayleigh additivity.** An  $n$ -dimensional grid is the Cartesian product of 1-D chains, and a product graph’s quadratic form separates exactly:  $\text{pgform}(x \otimes y) = \text{gform}_G(x) \|y\|^2 + \|x\|^2 \text{gform}_H(y)$  (`form_separates`), with the analogous exact identity for the norm (`norm_separates`). Consequently a Rayleigh pair on each factor lifts to a Rayleigh pair on the product with eigenvalues adding (`rayleigh_additive`) — pure  $\mathbb{Q}$  combinatorics, no eigenvalue theory or continuum limit invoked.

**Step 2 (Tier-1, `sigforall.dec+funext` only, confirmed no classic leak): the flat multi-axis readout.** Reusing this repository’s own 1-D symmetric-second-difference machinery (`tends0`, verbatim from the audited core) with two new closure lemmas (additivity, scaling), `weighted_two_axis_readout` shows a weighted sum of per-axis second-difference quotients converges to the same weighted sum of the per-axis limits. `box_2d_readout` (weights  $-1, 1$ ) reproduces this repository’s own `lorentz_box` construction as an instance — confirming the extension is consistent with the existing kernel, not a divergent reformulation. `laplacian_2d_readout/laplacian_3d_readout` give the flat  $n$ -D Laplacian.

**Step 3 (Tier-1, same axiom profile): the variable-coefficient case.** The flux-form stencil  $a(x+h/2)(f(x+h)-f(x)) - a(x-h/2)(f(x)-f(x-h))$ , divided by  $h^2$ , is shown (`weighted_second_difference_limit`) to converge to  $a(x)f''(x) + a'(x)f'(x) = (af')'(x)$  — the divergence-form operator emerges as a readout of the discrete flux stencil, not assumed. `constant_coefficient_instance` confirms consistency with Step 2.

**Step 4 (Tier-0): the off-diagonal dominance criterion, sharp both ways.** An anisotropic (cross-term-bearing) quadratic form is representable as a nonnegative-weighted graph form — i.e. as a genuine graph Laplacian — if and only if it is diagonally dominant (`repr_pos_iff_dominant`, `repr_neg_iff_dominant`). Sufficiency is a direct decomposition; necessity is extracted by evaluating any hypothetical representation at three probe vectors, forcing the cross-edge weight to a specific value that goes negative exactly when dominance fails — a genuine obstruction, not an artifact of one decomposition choice.

**What this closes and what remains open, by design.** Steps 1–4 together give the diagonal-metric Laplace–Beltrami operator class  $\sum_i \partial_i(a_i(x)\partial_i u)$  — covering Schwarzschild-radial, FRW, and essentially every metric weak-field tests use — with a sharp cri-

terion for when off-diagonal terms admit a graph representation at all. What remains open, stated at its narrowest: spectral convergence for a fully general (non-dominant) anisotropic metric, and general-metric convergence on a genuine manifold. The latter is a known, substantial result in the literature (Burago–Ivanov–Kurylev, 2014) [15]; it is cited here as prior literature, not mechanized, and is not claimed as this work’s contribution.

## 14 A Discrete Disk-Before-Lock Analog

A discrete, graph-native analog of an unrelated preprint’s continuum claim (“Disk Before Mass,” a PDE result stated with a provenance map but no worked derivation, hence **Dr** tier and not relied upon here) is built from scratch using only this repository’s own retention machinery. Under an explicit *anisotropy hypothesis* — edges classified “perpendicular” or “parallel,” with the perpendicular class’s benefit always strictly below its strain, and the parallel class’s always at or above — a perpendicular candidate edge is shown to strictly worsen the retention functional while a parallel candidate never does, from any current graph state, unconditionally. This is an immediate, per-decision fact, not an asymptotic claim about many growth steps: **InfoDiskBeforeLock.v** (5 theorems, **Th.coqc**). The anisotropy hypothesis itself is disclosed as input data, not derived.

## 15 Closing the Seed-to-Ceiling Chain

A separate card (Supplement) tracks a chain from the seed’s own retention criterion through curvature to an internal spectral ceiling. Two remaining gaps in that chain are closed here. **InfoGrowthFold.v** (5 theorems, **Th.coqc**) generalizes the single-edge curvature-shift law of §10 to an arbitrary *list* of added edges via an exact, order-independent closed form (**forman\_fold**), and shows retention only ever bends curvature in one direction regardless of growth order. **InfoCeilingMonotone.v** (6 theorems, **Th.coqc**) shows growth never decreases the quadratic form a Rayleigh quotient is built from, that every exact Rayleigh level achieved before growth is still dominated after growth, and that the degree data feeding the spectral ceiling of §10 only grows — explicitly disclosing what is *not* claimed (individual eigenvalue monotonicity/interlacing, which needs spectral theory not invoked here). Together these make the seed-through- ceiling portion of that chain gap-free up to one disclosed ansatz

and one pre-existing, direction-undecidable axiom relating an internal oscillation period to mass — neither of which is discharged by this section, and neither of which is claimed to be.

## 16 Algebraic Bridges: Curvature Symmetry, Rotation, and a Pentagon Spectrum

Three results connect the curvature/growth track to a separately-developed algebraic track (complex roots of unity, the golden ratio) via concrete shared objects, not analogy. **InfoCurvatureNoether.v** (3 theorems, **Th.coqc**) proves Forman curvature is invariant under any graph automorphism with a correspondingly permuted edge list, extending an existing Noether-style invariance result (proved there only for the Laplacian-built quadratic form) to curvature specifically for the first time; a concrete instance reuses an existing order-6 rotation witness, with the honest caveat that this particular witness graph is regular, so its curvature is trivially constant there — the general theorem is what matters for non-regular graphs. **InfoModeRotation.v** (8 theorems, **Th.coqc**) shows the mother equation’s own leapfrog/CFL stepper matrix, at the specific stability-window parameter value where its trace is zero, is *literally* the standard matrix representation of multiplication by  $i$  (verified: the same squares-to-negation, period-4 identity already known for  $i$ ), and that two further window points give periods 6 and 3 — instances of the crystallographic restriction theorem, cited as prior literature for the general claim (only  $n \in \{1, 2, 3, 4, 6\}$  are reachable by a rational stepper parameter), not re-derived here. **InfoPentagonSpectrum.v** (5 theorems, **Th.coqc**) proves the 5-cycle graph Laplacian satisfies an exact cubic operator identity whose nonzero-eigenvalue factor has the same defining-relation shape as the golden ratio’s own characterizing equation (both are minimal polynomials of elements of  $\mathbb{Q}(\sqrt{5})$ ), and exhibits all four distinct eigenvalues of the 6-cycle (the same graph used as the automorphism witness above) via explicit rational eigenvectors — the 6-cycle is, in this precise sense, the last cycle whose full spectrum stays rational; the 5-cycle is the first whose spectrum leaves  $\mathbb{Q}$ .

## 17 A Graph-Native Boundary-Energy Fact and a Degree-Curvature Bound

Two small, self-contained results, explicitly not claimed as more than what they state. **InfoAreaLaw.v** (3 theorems, **Th.coqc**) proves that if a field is constant across a region of a graph, every edge strictly interior to that

region contributes exactly zero to the quadratic energy form — all of a uniform region’s energy comes from its boundary. This shares a qualitative flavor with the Bekenstein–Hawking holographic bound (“boundary determines the budget, not the interior”) but is not that bound and makes no claim about entropy, metric area, or general relativity; sharing a qualitative flavor is not sharing a theorem. `InfoDegreeFromCurvature.v` (2 theorems, `Th_coqc`) proves a node’s degree is bounded by 4 minus the local Forman curvature of any incident edge, and, given a curvature floor over a graph, that the maximum degree is bounded by 4 minus that floor — a purely combinatorial fact usable to feed the spectral ceiling of §10 with a curvature-derived value. Stated plainly: given the file’s own definition `forman(e) := 4 - deg(u) - deg(v)`, the local bound is an immediate one-line rearrangement (equivalent to non-negativity of degree); the contribution is packaging this as a directly-usable input to the spectral ceiling, not a deep new fact about curvature.

## 18 General Tensor Reconstruction and a Conditional Curvature Bridge

`InfoTensorFrame.v` (7 theorems, `Th_coqc`, division-free over  $\mathbb{Q}$ ) proves a general, dimension-independent linear-algebra fact: a symmetric bilinear form on an  $n$ -dimensional space is fully and uniquely reconstructible from its  $n$  axis (diagonal) evaluations plus its  $\binom{n}{2}$  face-diagonal evaluations, via the polarization identity — the counting identity  $2n + n(n - 1) = n(n + 1)$  confirms no slack in any dimension, and in 4D, 4 axes + 6 face-diagonals = 10 matches a rank-2 symmetric tensor’s component count exactly. This is stated plainly as an *abstract* linear-algebra fact, deliberately not identified with general relativity’s metric or stress-energy tensor: no metric signature, no coordinate transformation law, and no field equation is claimed or used here, and the fact holds for a symmetric bilinear form on *any*  $n$ -dimensional  $\mathbb{Q}$ -vector space, gravitational or not. It also does not by itself establish that any graph actually used elsewhere in this work contains face-diagonal edges (the  $n$ -D ladder of §13 uses axis-aligned edges only).

`InfoStrainTensorBridge.v` (11 theorems, `Th_coqc`) connects this to curvature on a stated condition: on a  $2 \times 2$  graph cell with a diagonal candidate edge, the strain on that edge equals the tensor-frame diagonal evaluation *exactly when the cell is affine* (no discrete Hessian cross term), and differs from it by an exact, fully disclosed defect term in general. Under that affine condition, the seed’s own retention criterion for the diagonal candidate is shown to be, verbatim, a price on a tensor-frame evaluation (`diagonal_retention_prices_`

`tensor`) — curvature and this tensor-reconstruction fact are connected by a real theorem for the affine case, not merely both being expressible on the same graph. This is *not* an unconditional closure; for a non-affine cell the disclosed defect term must be carried. The same file separately proves that any graph’s quadratic form splits exactly into inside, outside, and cut contributions against any region (`gform_screen_partition`, unconditional), and, under one further disclosed lens (benefit per retained distinction taken proportional to a temperature-like parameter), that the retention criterion becomes a discrete Clausius-type inequality, both directions exact. None of this constitutes or is claimed to constitute Jacobson’s thermodynamic derivation of the Einstein equation [18]; it is a small, honestly-tiered fragment sharing a qualitative shape with one step of that program, explicitly not the program itself.

## 19 Momentum Optimizer Theory Coincides With the Kernel’s Energy Theory on a Quadratic Mode

`InfoOptimizerWindow.v` (10 theorems, `Th_coqc`, division-free over  $\mathbb{Q}$ ) shows the classical heavy-ball/momentum iteration on a quadratic mode is, under an exact reparametrization, the same dissipative two-step recurrence already proved for the mother equation’s own leapfrog stepper in §10. The reparametrized Lyapunov functional decrements by exactly one nonnegative square per step; a momentum dichotomy (never increases below a threshold, exactly conserved at the threshold, never decreases above it) is proved as a theorem; and an a-priori bound on every iterate is obtained from initial data alone, with no spectral computation or run-time tuning, inside the classical stability window. Divergence outside that window, convergence rates, and anything about non-quadratic objectives are explicitly not claimed.

## 20 Reviewer Attack Response

This section pre-empts questions an adversarial referee will ask, following an independent technical review of an earlier draft.

*Is  $E^2 M = \hbar^2 K \lambda$  really quantum mechanics?* No, not non-relativistic Schrödinger mechanics (correctly flagged: that theory is first-order in  $E$ ). It is a relativistic (Klein–Gordon-family) dispersion, and §6 proves this is exact, not a relabeling.

*Is the Hückel match a real validation?* It is a consistency demonstration of graph-topology sensitivity (confirmed



via a non-aromatic control), not an independent confirmation of a fitted physical constant (§4).

*Is Eq. 6 really “frame covariance”?* It is invariance under edge-list relabeling, a bookkeeping symmetry, not physical boost covariance; the latter (Eq. 7’s invariance under the imported boost formula) is stated and proved separately and is not conflated with it here.

*Is sgn in Eq. 4 actually derived from the causal order  $\prec$ ?* No — it is a free parameter in the source theorems. This manuscript states that explicitly (§5) rather than implying otherwise, and lists deriving it as open work (§24).

*Is the curvature–dissipation link (Eq. 12) non-trivial?* It is an exact algebraic identity, but shallow: it holds because both quantities are stated in terms of degree, not because curvature causes a dissipation requirement. This is disclosed, not obscured.

*Are all citations independently checkable?* Every citation in §25 was checked against standard physics/mathematics references; two citations flagged as unresolved in an internal draft (an anonymous arXiv preprint and a mismatched preprint server) have been removed from this version pending independent verification, rather than retained on trust.

*Does the frequency ceiling (Eq. 15) claim a physical Planck-scale minimal length?* No. It is a structural/discretization bound: a property of  $L_R$ ’s finite maximum degree. It states *that* a ceiling exists and *exactly* what sets it ( $d_{\max}$ ,  $M$ ,  $K$ ), not a numerical value for any particular physical system.

*Is the causal-signature closure (§10) actually complete?* No, and the manuscript says so directly: the construction is proved for an arbitrary relation with genuine incomparable pairs, demonstrated on a constructed example, but this repository’s own causal order is a total order and cannot host one. This is reported as a partial result with a named obstruction, not a closed gap.

*Does the graph-growth result (§11) address the Hubble tension or any cosmological discrepancy?* No, explicitly. That is a general-relativistic problem and §7 establishes no general relativity is derived here. The growth result is offered only as a native discrete analog of expansion, with an honestly bounded scope (power-law dilution only; no de Sitter behavior found; topology-dependent, not universal).

*Does the boundary-energy fact (§17) establish a holographic entropy bound?* No. It shares a qualitative flavor with the Bekenstein–Hawking bound (boundary determines the budget, not the interior) but proves a purely combinatorial fact about a quadratic energy form on a graph — no entropy, no metric area, and no general-relativistic structure appears in the statement or the proof.

*Does §18’s tensor-reconstruction fact establish anything about physical tensors (metric, stress-energy), and is the curvature bridge to it unconditional?* No on

both counts. The reconstruction fact is an abstract linear-algebra identity holding for any symmetric bilinear form on any  $n$ -dimensional  $\mathbb{Q}$ -vector space; it is not applied to, or claimed to be, general relativity’s metric or stress-energy tensor. The curvature bridge (`diagonal_retention_prices_tensor`) holds only on the stated affine-cell hypothesis; for a non-affine cell an explicitly disclosed defect term must be carried, and this is not an unconditional closure of any kind.

## 21 Novelty Audit

The individual mathematics throughout — Gershgorin/Rayleigh bounds, leapfrog invariants, discrete Green’s identities, Forman curvature, lattice Noether theorems, variational integrators, Kronecker sums, diagonal-dominance criteria — is classical or already in the literature, and this repository’s own discrete-substrate-physics framing sits in a crowded field (causal sets, the Wolfram Physics Project, Regge calculus, loop quantum gravity). A dedicated literature check (2026-07-05) on the five specific structural identifications introduced in §12–11 found no close prior art for three of them (the curvature-affine horizon threshold; the horizon dichotomy proved rather than stipulated; an epistemic-axiom-to-energy-floor bound, not mechanized here), one close but distinct prior construction requiring explicit citation (the double matter/geometry stationarity, §12, cf. Trugenberger’s Combinatorial Quantum Gravity [16]), and one instance of well-established qualitative folklore given a specific proved form here for the first time as far as this check found (the order-collapse theorem, §11, cf. causal set theory [17]).

**Scope limit, stated plainly.** This literature check covers only §12–11. The eleven files and six sections added later (§14–19) have NOT been through an equivalent dedicated check and their novelty is therefore *unaudited*: several of their individual ingredients are flagged in-line as classical or adjacent to known results (the crystallographic restriction theorem cited as prior literature in §16; the Bekenstein–Hawking qualitative resemblance disclaimed in §17; the polarization identity underlying §18), but no systematic search for close prior art was performed for these sections as a whole.

The claimed contribution is therefore narrower than “new physics” and is stated at that level: a single graph operator carrying genuine (not aliased) derivations — not results sharing a vocabulary, but a shared equation — across dispersion, causal structure, conservation, symmetry, curvature, and variational structure simultaneously, machine-checked and axiom-free end to end, with a tier system that separates theorem from stance at every step. This is a claim about *method and integration*, not about any single physical prediction; the one structural prediction native to this framework is the fre-

quency/UV ceiling of §10 (Eq. 15), which has not been tested against any experiment.

## 22 Main-Text vs. Supplement Boundary

This manuscript is self-contained for its theorem-level claims: every stated equation carries its exact source file and tier, and §23 gives the exact commands to independently re-check every one. The manuscript is responsible for the mother equation, the four branch results (§4–8), the unification result (§6), the quasinormal-mode bridge (§9), the strengthening campaign (§10), graph growth (§11), the action-stationarity and curvature-horizon structure (§12), the  $n$ -D spectral ladder (§13), the disk-before-lock analog (§14), the seed-to-ceiling closure (§15), the algebraic bridges (§16), the boundary-energy fact and degree-curvature bound (§17), the tensor-reconstruction and conditional curvature bridge (§18), the momentum-optimizer identification (§19), the reviewer attack response (§20), and the reproduction commands.

The companion supplement (SUPPLEMENT.md) is responsible for material this manuscript summarizes but does not reproduce in full: the complete dependency DAG in both diagram and prose form; the full log of all eight attempts to derive general relativity, including the ones only briefly mentioned in §7; the complete novelty audit against prior discrete-substrate literature (§21 gives only its conclusion); and the per-citation verification notes recording how each reference in §25 was checked, including two citations present in an earlier internal draft and removed after they could not be independently verified (§20). A reader who wants the theorem statements and can reproduce them needs only this manuscript; a reader who wants the full argument trail, including dead ends, needs the supplement as well.

## 23 Reproducibility Protocol

1. `git clone` this repository.
2. `make verify` — compiles all 35 `formal/*.v` files via `coqc` in dependency order and runs `Print Assumptions` on every named theorem (188 total), printing a tier report (`Th.coqc / +reals` per file).
3. `python3 scripts/verify-quantum-gravity-root-bridge.py` — reproduces the quasinormal-mode convergence table (Eq. 14).
4. Expected result: all Coq files compile with exit code 0; the tier report matches Table 1; the QNM script’s  $N = 3200$  row matches Eq. 14 to the printed precision ( $N = 1600$  is already close,  $0.4838 - 0.0958i$ , but the digits quoted in Eq. 14 are the  $N = 3200$  row).

## 24 Open Problems

(1) Construct a genuinely non-total (partial, multi-dimensional) causal order on this repository’s graphs, so that the sign-construction theorem of §10 produces a non-degenerate indefinite signature on the repository’s *own* causal structure, not only on a constructed example. (2) Extend the quasinormal-mode bridge (§9) to spin-2 perturbations. (3) Determine whether Ollivier–Ricci curvature [7] offers a non-shallow alternative to Eq. 12’s exact-but-shallow link. (4) Does the dilution law of §11 admit a principled selection rule for which growth topologies are physically relevant, or is topology-dependence the final answer? (5) Does §10’s Noether pairing admit a principled dissipative correction, or is exact conservation strictly a conservative-sector fact? (6) Discharge, or find a counterexample to, the pre-existing, direction-undecidable axiom relating an internal oscillation period to mass that §15 leaves untouched — the seed-to-ceiling chain is gap-free only up to this axiom. (7) Derive, rather than disclose as input data, the anisotropy hypothesis of §14 (which edges count as “perpendicular” vs. “parallel” and why) from a more primitive graph property, so the disk-before-lock result no longer depends on an externally-supplied classification.

## 25 Conclusion

One graph-Laplacian root equation genuinely derives, and genuinely unifies, a quantum dispersion relation and a special-relativistic wave operator. It honestly does not derive general relativity, for a stated philosophical reason rather than a numerical shortfall, and offers in its place a native discrete curvature readout with an exact, if shallow, link to an independently-proven stability result. Responding to adversarial review, six further theorems promote a frequency ceiling, an energy-balance law, a first-order unitary-like skeleton, a causal sign construction (partially closed, honestly), and a discrete Noether law from informal stances to machine-checked facts; a seventh shows that graph growth is a native, non-continuum, honestly-bounded analog of expansion, explicitly not a claim about any physical cosmological discrepancy. Two further results prove the mother equation is an exact stationarity readout on both its field and its geometry, and compose the graph’s own curvature into an exact horizon threshold and dichotomy — reported at full strength for what they establish, and explicitly not as evidence of narrowing any gap toward general relativity’s field equations, since variational-structure-sharing is generic to essentially all physical theories. A final four-step ladder closes the flat and diagonal-metric cases of this framework’s  $n$ -dimensional contin-

uum limit, tagging each step at its true tier and narrowing the remaining general-metric case to a single, named, literature-cited gap. Eleven further files extend the reach without ever advancing a claim past its evidence: a discrete anisotropy result immediate at every decision point, not asymptotic; two files closing a seed-to-spectral-ceiling chain up to one disclosed ansatz and one pre-existing, direction-undecidable axiom, neither discharged here; three files bridging curvature and growth to a separately-built algebraic track via concrete shared objects (an automorphism-invariance extension, a literal identity between the mother equation’s own stepper matrix and the standard  $i$ -rotation matrix, and a pentagon-graph spectrum landing in the same field as the golden ratio); a graph-native boundary-energy fact and a degree-curvature bound, each explicitly scoped against the physical results (holographic entropy, general relativity) they share only a qualitative flavor with, not a theorem; a general tensor- reconstruction fact and a curvature bridge to it that is conditional on one named hypothesis, not unconditional; and a momentum-optimizer identification inheriting the mother equation’s own energy theory by exact reparametrization. Every claim above carries its tier and its exact reproduction command.

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This work was developed through a human–AI co-thinking workflow, including one round of independent adversarial technical review external to the author’s own assessment. Any remaining errors are the author’s own.

## Author Declarations

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**Data and Code Availability.** All Coq sources, the Python reproduction script, and this manuscript’s source are available in the accompanying repository.

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